

## LAB 6: Interaction Diagrams: Sequence Diagram

### Objective

The objective of this lab is to practice sequence diagrams.

### Theoretical Description

UML sequence diagrams model the flow of logic within your system in a visual manner, enabling you both to document and validate your logic, and are commonly used for both analysis and design purposes.

Sequence diagrams are typically used to model:

1. Usage scenarios. A usage scenario is a description of a potential way your system is used. The logic of a usage scenario may be part of a use case, perhaps an alternate course. It may also be one entire pass through a use case, such as the logic described by the basic course of action or a portion of the basic course of action, plus one or more alternate scenarios. The logic of a usage scenario may also be a pass through the logic contained in several use cases. For example, a student enrolls in the university, and then immediately enrolls in three seminars.
2. The logic of methods. Sequence diagrams can be used to explore the logic of a complex operation, function, or procedure. One way to think of sequence diagrams, particularly highly detailed diagrams, is as visual object code.
3. The logic of services. A service is effectively a high-level method, often one that can be invoked by a wide variety of clients. This includes web-services as well as business transactions implemented by a variety of technologies such as CICS/COBOL or CORBA-compliant object request brokers (ORBs).

### Case Study: ATM Cash Withdrawal

#### 1. Brief Description

This use case describes how the Bank Customer uses the ATM to withdraw money to his/her bank account.

#### 2. Actors

- Bank Customer
- Bank

#### 3. Preconditions

- There is an active network connection to the Bank.  
The ATM has cash available.

#### 4. Basic Flow of Events

- a) The use case begins when Bank Customer inserts their Bank Card.
- b) Use Case: Validate User is performed.

- c) The ATM displays the different alternatives that are available on this unit. In this case the Bank Customer always selects “Withdraw Cash”.
- d) The ATM prompts for an account.
- e) The Bank Customer selects an account.
- f) The ATM prompts for an amount.
- g) The Bank Customer enters an amount.
- h) Card ID, PIN, amount and account is sent to Bank as a transaction. The Bank Consortium replies with a go/no go reply telling if the transaction is ok.
- i) Then money is dispensed.
- j) The Bank Card is returned.
- k) The receipt is printed.
- l) The use case ends successfully.

## 5. Alternative Flows

### 5.1 Invalid User

If in step 2 of the basic flow Bank Customer the use case: Validate User does not complete successfully, then the use case ends with a failure condition

### 5.2 Wrong account

If in step 8 of the basic flow the account selected by the Bank Customer is not associated with this bank card, then

1. The ATM shall display the message “Invalid Account – please try again”.
2. The use case resumes at step 4.

### 5.3 Wrong amount

If in step 7 in the basic flow, the Bank Customer enters an amount that can't be 'created' with the kind of in the ATM, then

1. The ATM shall display the message indicating that the amount must be a multiple of the bills on hand, and ask the Bank Customer to reenter the amount.
2. The use case resumes at step 7.

### 5.4 Amount Exceeds Withdrawal Limit

If in step 7 in the basic flow, the Bank Customer enters an amount that exceeds the withdrawal limit (See Special Requirement WC-2 for maximum amount), then

1. The ATM shall display a warning message, and ask the Bank Customer to reenter the amount
2. The use case resumes at step 7

### 5.5 Amount Exceeds Daily Withdrawal Limit

If in step 8 in the basic flow, the Bank response indicates the daily withdrawal limit has been exceeded (this is determined by the Bank and depends upon the specific account), then

1. The ATM shall display a warning message, and ask the Bank Customer to reenter the amount.
2. The use case resumes at step 7.

## 5.6 Insufficient Cash

If in step 7 in the basic flow, the Bank Customer enters an amount that exceeds the amount of cash available in the ATM, then

1. The ATM will display a warning message, and ask the Bank Customer to reenter the amount.
2. The use case resumes at step 7.

## 5.7 No Response from Bank

If in step 8 of the basic there is no response from the Bank within 3 seconds, then

1. The ATM will re-try, up to three times.
2. If there is still no response from the Bank, the ATM shall display the message “Network unavailable – try again later”.
3. The ATM shall return the card.
4. The ATM shall indicate that it is “Closed”.
5. The use case ends with a failure condition.

## 5.8 Money Not Removed

If in step 9 of the basic flow the money is not removed from the machine within 15 seconds, then

1. The ATM shall issue a warning sound and display the message “Please remove cash”.
2. If there is still no response from the Bank Customer within 15 seconds the ATM will retract the money and note the failure in the log.
3. The use case end with a failure condition.

## 5.9 Quit

If at point prior to step 8 in the basic flow the Bank Customer selects Quit, then

1. The ATM shall print a receipt indicating the transaction was cancelled.
2. The ATM shall return the card.
3. The use case ends.

## 6. Key Scenarios

### 6.1 No Response from Bank

## 7. Post-conditions

### 7.1 Successful Completion

The user has received their cash and the internal logs have been updated.

### 7.2 Failure Condition

The logs have been updated accordingly.

## 8. Special Requirements

- The ATM shall dispense cash in multiples of \$20.
- The maximum individual withdrawal is \$500.
- [SpReq:WC-1] The ATM shall keep a log, including date and time, of all complete and incomplete transactions with the Bank.

## NOTE

1. Start MS Visio. Select Softwares and Databases category. Select UML Model Diagrams.
2. Select Sequence Diagrams.
3. An actor notation can be added from the use-case diagram section and 'Life Line' object can be used with it instead of an 'Object Life Line'.

## Lab Task

- Draw the sequence diagram for the above mention case study